Prioritizing Education: The Macao Experience Since the Millennium*

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Abstract

Macao is globally recognized as a high-performing system according to the latest releases of PISA results. Such recognition comprises two aspects: 1) average performance of all Macao fifteen-year-olds was among the highest in the world; and 2) there is a considerably weak link between socioeconomic status and learning outcomes in Macao compared with other participating economies. This chapter examines the systemic effort behind these phenomena. A brief history characterizes the special administrative region in two periods. Before 1999, the Macao system had long been criticized for its uncoordinated structure originating from the laissez faire regime under the historical Portuguese colonial bureaucracy. After 1999 and to this date, Macao went through a gradual transition from a loosely coupled system to a re-coupling one. The government's effort is most evident in elaborating an extant legal framework to regulate schools and teachers. Riding a booming economy, the government consistently prioritized educational development. Its expanding and targeted fiscal policy allowed Macao schools to reduce class sizes while maintaining an engaging variety of extra-curricular activities. Due to the absence of a central, standardized examination, the government also made use of international large-scale assessments to monitor school performances. The authors argue that both high- and low-performing students benefited from these policies. The authors close the chapter by discussing the challenges ahead.

Introduction

Historically, the education system in Macao received little attention from the international community. As a tiny peninsula with two island districts, known as the Macao Peninsula with the island of Taipa and the reclaimed land of Cotai, Macao was sparsely populated. Although its geographic size has gradually increased as a result of continued land reclamation, from 15.5 km^2 in 1980 to 32.9 km^2 in 2020, the territory remained small compared with neighboring cities such as Hong Kong and Singapore. So was its population. It grew from 248,636 in 1980 to 679,600 in 2020 (Macao Yearbook Editorial Committee, 2020; Repartição dos Serviços de Estatística, 1981).

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The small size, both geographically and demographically, shaped the local organization of education. On the one hand, all schools were clustered closely with one another (Thomas & Postlethwaite, 1983). According to the most recent government statistics, there are 77 schools in Macao as of 2022, with 120 sections (i.e., a school can have separate kindergarten, primary, and secondary sections). Among the 77, 10 are public schools and 67 are private schools. A survey of school spaces and environment at the non-tertiary level showed that overall, there was a school nearby every 500 meters. Eighty seven percent of schools were located in the Macao Peninsula, where the distance between any two schools was no more than 400 meters. Regarding school size, 50% schools had fewer than 500 students and over 90% fewer than 2,000 students (Direcção dos Serviços de Educação e Juventude [DSEJ], 2007). The geographic proximity made it possible for a simple two-level hierarchical structure, with a department of education at the center and an array of local schools attached to it. On the other hand, the small market impeded the development of an indigenous system (Bray & Hui, 1989). Local school systems were directly imported from neighboring regions. Apart from being a place in miniature, as an uncoordinated and fragmented education system, Macao hardly attracted any scholarly attention. Under the formerly Portuguese colonial bureaucracy, public schools were open only to the Portuguese-speaking population. Chinese students, comprising the majority of the schooling population, attended private schools run by the diocese, and charity and grassroots civil organizations. Without funding from the colonial government, these private schools were left on their own. As the supply of students was unstable, most schools faced financial deficits and ran in poor conditions. Since private schools did not rely on public support, the government seldom got involved in private schooling, and these schools held great independence and autonomy in organizing learning activities within schools. As a result, private schools varied in terms of curricular content, instructional medium, and teacher quality (Adamson & Titus, 2005).

In the past decade, Macao has garnered increasing research interest. Such interest was due to Macao's improving standing in successive rounds of international large-scale assessments, where Macao claimed the titled of a high-performing system. Yet, most studies simply included Macao as one of the participating economies and examined the role of system-level indicators in facilitating learning without taking more in-depth context into consideration (e.g., Chiu et al., 2012; Ning, 2019). The context-rich studies that did exist usually deployed Macao as a comparison with Hong Kong in the 1990s and 2000s (e.g., Adamson & Titus, 2005). This line of research provided structured and detailed comparisons before and shortly after each system's handover to mainland China. However, after examining the literature, the authors concluded there have not been any systematic studies on Macao using quantitative and qualitative evidence in the recent decade. The authors believe that this is a missed opportunity. Macao is currently gradually transitioning towards a cohesive and integral education system. It is high time to look back.

In this chapter, the authors delineate the changing landscapes in the educational sector after the millennium. Since then, Macao has achieved noticeable educational progress in academic domains and extra-curricular activities. It is important to acknowledge that this period resembled a dynamic process whereby the education system was shaped by specific historical and political settings (i.e., Elder, 1978). In addition to the historical complexity, the authors wish to emphasize the importance of equity-oriented policies in guiding the systemic improvements. At an age when many countries shifted towards academic competition and held schools accountable for standardized learning outcomes (Baker & LeTendre, 2005; Yurkofsky, 2020), Macao buffered students and schools from such external demands by investing in a quality school system and espousing students' diverse potential. The authors argue that the strong emphasis on equity, respect to institutional diversity, and the corresponding financial support were essential for its educational success. The authors close the chapter by discussing the implications and challenges ahead.

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Changing landscapes

Unwavering commitment to education

The education system left by the colonial government was an unenviable legacy. The Portuguese colonial administration took a laissez faire attitude toward educational affairs during its rule. Its inattention to educational development was embodied in its non-interventionist approach toward private schooling organized by Chinese communities and its lopsided resource-allocation approaches. The administration channeled most educational resources to the few extant public schools. In those schools, the academic structure was uniformly aligned with the Portuguese scheme. In a departure from the Portuguese system, private schools could decide their own schemes since the administration was rarely involved in their operations. Despite the fact that most students attended Chinese-run private schools, a disproportionate amount of government funding went to public schools. According to an assessment survey conducted in 1986, local officials disclosed that 70% of educational funds went to the 10 public schools while the remaining 69 schools received a meager 30% (Yee, 1990). Educational development did not enter the public policy agenda until the Sino-Portuguese declaration in 1987. Despite the colonial government's intention to enact sweeping reforms, there were few changes due to a lack of preparation and the limited time available before the handover (1999).

Since its establishment as a special administrative region of China, Macao has consistently endeavored to improve the education sector. The Education and Youth Affairs Bureau (Portuguese: Direcção dos Serviços de Educação e Juventude, or DSEJ as commonly known) is the department in charge of the development and organization of non-tertiary education. Although the DSEJ was restructured to the Department of Non-tertiary Education under the new Education and Youth Development Bureau (Portuguese: Direcção dos Serviços de Educação e de Desenvolvimento da Juventude, or DSEDJ) in 2021, which has a much larger jurisdiction, these two institutions are not differentiated for the purpose of this chapter. The two supporting units, the Student Welfare Fund and the Education Development Fund, are responsible for providing need-based aid to students and financing school-based development projects. Together, these were the three main channels of school funding.

Table 1 shows public investment in non-tertiary education since 1999. Two things stand out. First, over time, a substantial amount of public funds has been continuously allocated to the non-tertiary education sector. Second, the scale of investment has grown year by year. In the first decade, from 1999 to 2009, public expenditure on non-tertiary education almost tripled. In the second decade (2009-2019), the same allocation reached 1 billion USD in 2019, seven times higher compared to the 1999 number. The compound growth rate during the above-mentioned period was over 10% annually. The proportion of non-tertiary educational expenditure to the entire government's spending has also increased, up from 6% in 1999 to 9.6% in 2019. These figures all suggest that the government has set a clear priority on educational development.

The public expenditure provided a foundation for Macao's free education provision. Previously, free education was limited to public schools. Since 1995/96 academic year, the free tuition provision has spread to private schools as well. Most of the private schools have joined the "free education school network." Private schools within this network can receive subsidies from the government and provide tuition-free schooling. In the 2018/19 academic year, an equivalent of 325 million USD (2,597 million MOP) was allocated to the implementation of free education. In the same year, free education provision covered 67,766 students, accounting for 87% of the total student body (DSEJ, 2020a). Students who studied outside of the network can apply for a school tuition fee subsidy, an additional nine percent of students (7,009) received that subsidy. Combining the two numbers,

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Table 1: Government expenditure in non-tertiary education since 1999

Year	Expenditure on non-tertiary education* (USD in millions)	Total public expenditure (USD in millions)	Percentage share of non-tertiary education	GDP per capital (USD)
1999	127	2,080	6.08%	15,304
2004	143	2,213	6.45%	23,474
2009	363	4,432	8.19%	40,034
2014	675	8,385	8.05%	88,148
2019	1,017	10,585	9.60%	82,613

Notes.
The data are from Education and Youth Affairs Bureau, Financial Services Bureau and Statistics and Census Service.
The exchange rate between USD and MOP is pegged at 1:8.
* Government expenditure in non-tertiary education included spending on (a) Education and Youth Affairs Bureau, (b) Student Welfare Fund, and (c) Education Development Fund.

over 95% of students have benefited from free or subsidized education.

To put Macao's investment in education into context, the authors compared the cumulative public expenditure (i.e., primary plus secondary education) per student aged from 6 to 15 with neighboring East Asia systems in Figure 1. The data was compiled from OECD's PISA project. With 149,234 USD in 2015, Macao tops the chart. The amount spent per student was 50% more compared to Hong Kong and Japan, and 32% more than South Korea. This number implies a strong emphasis on education from public support. In fact, it was highlighted in Macao's Ten-Year Plan for the Development of Non-Tertiary Education (2011-2020) (Macao SAR Government, 2011) that the priority placed on education development would remain on the government's agenda as an important part of long-term policy. One of the plan's key measures was to increase annual investment in non-tertiary education based on economic development and financial stability. Riding on the booming economic growth covering the same time period, Macao has witnessed radically expanding government funding in education, revealing an unwavering priority for education.

The school finance statistics are not a onetime façade. The commitment to education is consistent. By reviewing the annual policy addresses from 2000 to 2020, the authors found that education development was consistently framed as a core strategy for the invigoration of Macao. Despite personnel changes and term shifts, the guiding principles and general policies for education have always stayed on the top policy agenda over the past two decades. In sum, educational development in Macao has been steadily supported by strong public financing.

Expanding access to quality education

Many resources went into expanding educational opportunities and improving the learning environment. Macao was the first in the Greater China region to provide 15 years of free education to its residents. Since the 2007/08 academic year, free education has covered all stages in non-tertiary education—three years of early childhood (age three to six), six years of primary school, three years of lower secondary school, and three years of upper secondary education. The government also ex-

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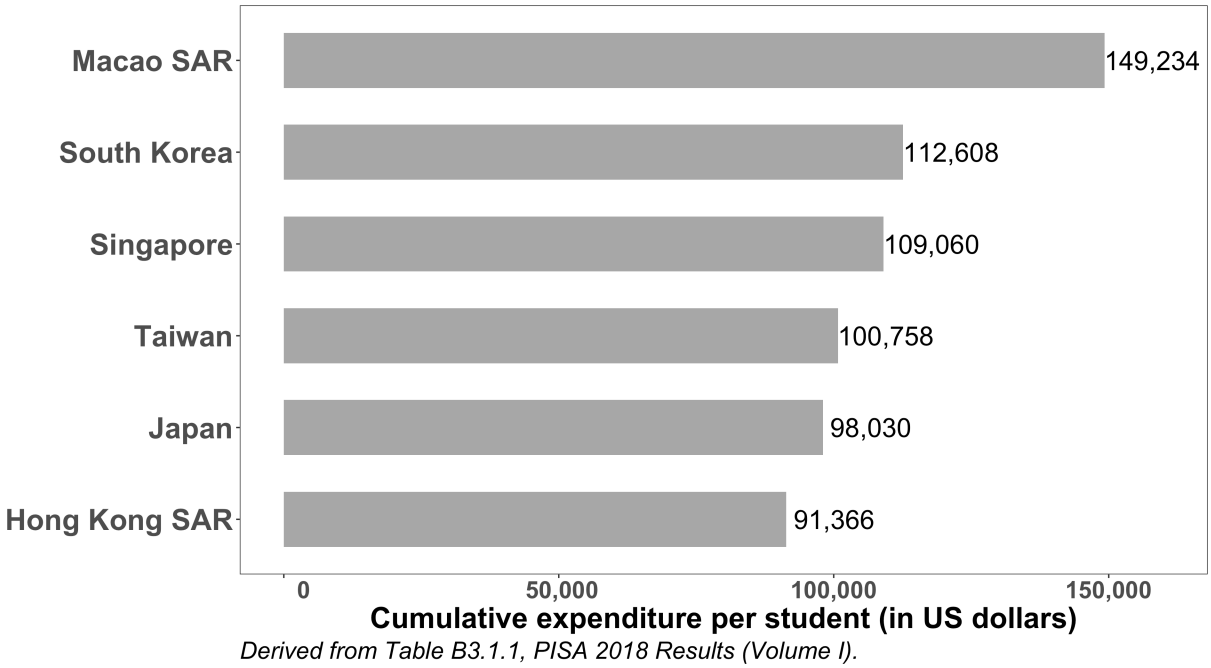

Figure 1: Cumulative expenditure per student aged from 6 to 15 (2015).

panded early childhood services by subsidizing the parents. Children as young as three months can attend a full-day nursery.

Continued investment in education translated into visible progress in Macao's schools. Improvements in the provision of education are most notably indicated by diminishing class sizes and improving compensation for teachers. Regarding the class sizes, the authors plot the annual figure by different levels of education in Figure 2 (black lines, scale marked by the labels on the right side of the y-axis). Available data allows us to track the data from 1996 all the way up to the most recent years available. The general trend indicated that class size shrank across every educational level. At the early childhood level, average class size decreased from 42 to 28 per class; at the primary level, from 47 to 30; at the lower secondary level, from 45 to 28; and at the upper secondary level, from 36 to 25.

The reasons behind such across-the-board decrease are twofold. First of all, Macao's birth rate fell sharply in the early 2000s, which resulted in a smaller schooling population. The birthing cohort had a size of 4,434 in 1998 (Census and Statistics Department of Macao, 1999). In subsequent years, it shrunk to 3,162 in 2002 and gradually climbed back to the same level only in 2007. Such a demographic change pushed the total school population to very low levels (grey lines, scale marked by the labels on the left side of the y-axis). Secondly, government policy encouraged schools to reduce class size by providing financial incentives. The free education allowance targets individual classes. If a class meets the size requirement of 25 to 35 students per class, then the school can receive an allowance for that class as a whole. The subsidy per class has increased across time, as shown in Figure 2 (represented by the bars). Driven by these forces, small class size was quickly adopted among schools. With a smaller classroom, teachers can devote more attention to each student's needs and students can have more opportunity to engage in their class (i.e., Finn et al., 2003).

Teacher policy has become a center piece in Macao's bid to improve education quality. The

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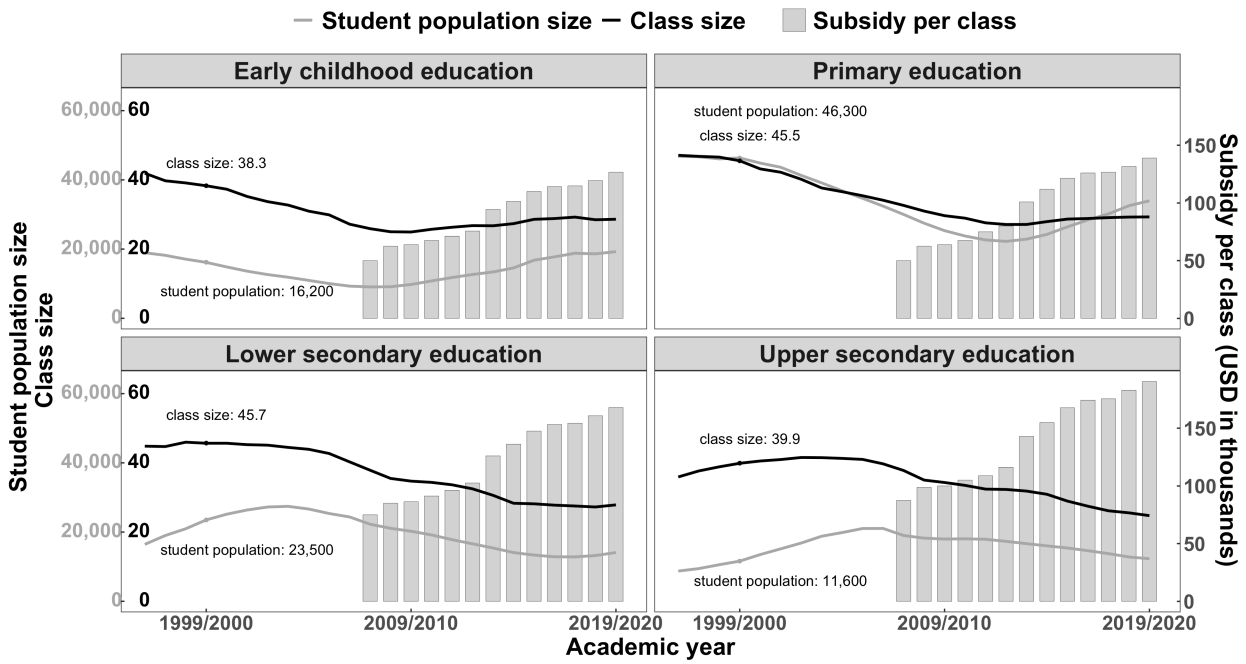

Figure 2: Trends in student population, class size and subsidy per class.

effort involves improving teacher compensation to attract talent and stabilize the labor market, as well as encourage teachers to pursue professional development opportunities. Macao is a late starter regarding teacher education. The first higher education institution for teacher education was established in 1989, today known as the Faculty of Education at the University of Macau. Before that, there was only one secondary school offering teacher training courses for primary education. Some teachers received their qualifications from neighboring regions such as mainland China, Taiwan or Hong Kong, yet many did not have any professional training. In 1985, only 22.4% of educational practitioners had received professional training (Lau, 2010). To enhance teacher professionalism, the government subsidized in-service teacher training. The majority of teacher education programs are now housed inside higher education institutions. These policies steadily raised qualification levels among Macao's teachers. According to DSEJ statistics, the proportion of teachers with teaching qualification has increased from 92.1%, 85.8%, and 72.1% in the 2010/11 academic year to 99.7%, 99.3%, and 93.5% in the 2018/19 academic year for early childhood, primary, and secondary education, respectively.

One important factor that allows schools to attract young talent is the improvement of remuneration. Teacher compensation was not competitive in the past. The report of a ten-year longitudinal survey on teacher attrition showed that there was a striking teacher salary variation among secondary schools during 1978-1988. In an extreme case, teachers in one private school earned only half as much as those in another and one third as much as those in public schools (Lau, 2010). Although the colonial government began to provide a monthly allowance of 500 MOP (around 62.5 USD) for private teachers in 1985, private high schools still suffered from a severe drain of teachers: teachers taught on average only three years at their current school. It was not until the introduction of Macao's first teacher law, the *System Framework for Private School Teaching Staff of Non-Tertiary Education* (Law No. 3/2012), in 2012, that the teacher labor market began to stabilize.

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The framework provided legal grounds for the adequate payment of teachers. With the government's continuing investment in the education sector, teachers' salaries have risen quickly. The starting salary for a secondary teacher in Macao was 97,925 USD annually in 2017, highest among countries and regions in East Asia. For comparison, the second highest pay was 62,057 USD in Hong Kong (OECD, 2019a). Being a teacher in Macao today means leading a decent life and a prosperous career, which helps to attract top high-school graduates to teacher education programs. According to annual reports of the survey on the study choices of senior high school graduates in Macao released by the DSEJ, a growing number of high school graduates pursued teacher education in tertiary institutions, the proportion of aspirated high school graduates has increased from 5.7% in the 2012/13 academic year to 10.8% in the 2019/20 academic year.

A rising professional profile led to more qualified individuals entering the teacher profession. According to DSEJ statistics, in the 1996/97 academic year, only 33.2% teachers owned a bachelor's degree or above. Yet in the 2020/21 academic year, 95.6% teachers had a bachelor's degree or above, and 17.6% had a master's or doctoral degree. With policy support for professional development and improved working conditions for teachers, the professionalism of in-service teachers has improved and pre-service teachers with higher qualifications have been recruited. Both contribute to the improved quality of instruction in the classroom.

Continuing effort to build an integrated system

The inconsistent curricula across schools, an unpleasant remnant of the past, were a key challenge confronted by the Macao government. Evaluation and revision of the curricular framework has been regarded as one of the major undertakings since 2000. The *Curriculum Framework for Formal Education of Local Education System* (Administrative Regulation No. 15/2014) stands out as one of the notable landmarks in the progress of curricular integration. In this localized framework, the general structure of the school curriculum was formulated, yet schools retained flexibility in developing school-based curricula to foster learning. The framework laid a strong emphasis on diverse development, concerning not only academic learning activities but also extra-curricular activities. Teachers were encouraged to pay attention to different students' learning needs and to provide situated learning assistance. The curricular framework is applicable to all schools following the local education system. Under the framework, teaching activities comprised six general areas: language and literature, mathematics, personal, social and humanities education, science and scientific technology, physical education and health, and art. Minimum amount of time spent on each learning area was specified, guaranteeing a threshold level of student involvement in educational activities.

In line with a localized curriculum framework are the *Requirements of Basic Academic Attainments* (Administrative Regulation No. 10/2015), which formulate the key competencies students are expected to acquire. The English title of this document is somewhat misleading. The authors think it is better to understand it as "competency requirement," not "attainment level." These competencies comprise not only the common 21st century skills such as creative problem solving and global perspective (i.e., Voogt & Roblin, 2012), but also broadly defined capabilities such as a sense of civic duty and appreciation of arts. To enrich student learning experience, a variety of free or subsidized extra-curricular activities are available for primary and secondary students. The extra-curricular activities include foreign language, music, art, sports, and STEM-based activities. For senior students, there are career advisors providing individual guidance and consultations about career planning. Schools were required to provide necessary assistance for students with learning difficulties and learning support for talented students to fulfill their potential. In this way, both low- and high-performing students could benefit from school-based activities. Noteworthily, even

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during the semester break in summer, students still have access to a wide range of sports and cultural activities launched by the DSEJ that aim to enrich their leisure and social experience. Furthermore, there is no high-stakes standardized examination in Macao. Colleges and universities can decide their own recruiting strategies according to the field of study. Examination scores are by no means the only admission criterion, students with merits in non-academic fields such as arts, sports or community service are also finding favor with colleges and universities. Results of annual surveys on study pathway showed that the proportion of high school graduates enrolled in higher education institutions has increased from 75.8% in 2000/01, to 85.9% in 2010/11, and up to 93% in the 2019/20 academic year, indicating a trend towards universal access to higher education (DSEJ, 2001, 2012, 2020b).

On the other hand, integration also takes the form of paying attention to students' special needs. Special education takes the form of dedicated special education schools for students with severe disabilities, as well as special education classrooms and inclusive classrooms inside mainstream schools. All three forms of special education are free to parents. Special education students receive free breakfast, lunch, and school bus service. They can also apply for financial assistance for auxiliary services. For immigrant students, specialized assistance and service such as language courses and consulting services are well-established to help them adapt to the new environment. By aligning various school practices and attending to different learning needs, Macao has developed into an increasingly integrated system.

Recent progress in educational achievement

Evidence from international assessments shows that these reforms and restructuring efforts paid off. To gauge students' performance, the Macao government requested in the early 2000s to use the OECD's popular Programme for International Student Assessment (PISA) as a barometer, since there is not any centralized testing regime at the primary or secondary level. In this section, the authors demonstrate that Macao students have made progress in educational achievement, not only in academic domains but also in non-academic areas.

PISA belongs to a family of international large-scale assessments conducted by the Organisation for Economic Co-operation and Development (OECD). It is carried out every three years among OECD countries and partner systems to evaluate 15-year-old students' ability to use their knowledge and skills to meet real-life challenges (OECD, 2017).

Mean Test Score

Due to Macao's small population, except for 2003 where a sample of 1250 was used, in all successive cycles, Macao has counted every 15-year-old in the PISA studies. No schools or school-aged students are excluded from PISA. Table 2 condenses past test results for Macao. The analysis performed here used public data files from PISA; the authors did not use any confidential sources.

Because the number of participating systems varied from cycle to cycle, it is not very meaningful to count the individual ranking. Instead, the authors used the more plausible "range of ranks" displayed in Table 2. Thus, one can interpret Macao students' reading performance in 2018 as ranked 3rd to 5th position in the 2018 PISA. This is because PISA ranking has always been controversial (Singer et al., 2018). As a quantitative exercise, the authors take into account the inherent uncertainty of a numeric test by calculating the range of possible ranking.

Table 2 shows that although consistently ranked at the upper half of performance distribution across years, Macao students always performed better in math and science as compared to reading. The ranking of Macao has moved forward since 2009 in all three test domains. Of the three, reading

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Table 2: Description of Macao's performance in PISA (2003-2018)

	No. of participating		Coverage rate	No. of participating economies	Range-of-rank estimates					
					Reading		Mathematics		Science	
Year	students	schools			Upper	Lower	Upper	Lower	Upper	Lower
2003	1250	39	100%	41	12	19	6	12	5	10
2006	4760	45	100%	57	18	22	7	11	15	20
2009	5952	45	99.9%	65	27	30	10	12	16	19
2012	5335	45	99.8%	65	13	22	6	8	13	17
2015	4476	45	99.9%	72	10	13	2	4	5	9
2018	3775	45	99.6%	79	3	5	3	4	3	3

Notes. Range-of-rank estimates are the 97.5% confidence interval for rankings among all participating economies. Since not only estimates of mean performance but also the rank estimates derived from them have some degree of uncertainty, the authors chose to report a confidence interval for ranks using one-tailed tests.

performance improved the most. Math and science performance has always remained at a very high level. In the latest 2018 cycle, Macao's performance in all three domains reached the global top five, even by conservative estimates.

It is worth noting that the Macao "sample" is an inclusive one. Every in-school student was counted. Unlike critics of some systems – such as the Shanghai sample which only allowed a selected sample into PISA (Carnoy, 2015) – Macao's schools are accessible to migrant/immigrant children, and even guest workers' children. The PISA student survey revealed that a significant share of the students were not born locally (that number is 31.2% in 2018). Therefore, Macao's achievement quite accurately reflected the learning outcome of the 15-year-olds in the territory.

A closer look at the students' performance distribution also revealed some uniqueness in the students' learning profile. The high mean test scores are reflected in two aspects: 1) there are fewer low-performing students; 2) the distribution is less spread out. In PISA, the Level-2 proficiency is identified as the threshold of proficiency that all children are supposed to acquire by the end of secondary education (OECD, 2019a, p. 89). Among all participating systems in 2018, Macao has the second largest proportion of students that have achieved this minimum proficiency level in all three assessment domains (89.2% in reading, 95% in math, and 94% in science), only behind mainland China (94.8%, 97.6%, and 97.9% respectively).

On the other hand, there were fewer high-performing students. Unlike the stereotypical high-performing countries where there is a large part of top students piling on the upper end (e.g., 7.3% in Singapore and 4.2% in mainland China), data from PISA 2018 show that only 2.1% of Macao students have achieved the highest proficiency level in reading (Level-6 or above 698.32 score points). In addition, the authors compare the average performance of top-performers (e.g., students at the 90th or 95th performance percentile) within each system. The mean performance at the 95th percentile in reading is 714 points in Singapore and 692 points in mainland China, both are statistically higher than Macao (670). This comparison suggests Macao's academic advantage fades as it moves to the higher end of the performance distribution. Similar results can be found in math and science as well (details see Annex B1, OECD, 2019a). Therefore, Macao's performance can be characterized as an overall high mean score and low dispersion. With a majority of students

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Table 3: Mean difference and boy-to-girl ratio in top performance percentiles, PISA 2018 (math).

	Mean performance			Boy-to-girl ratio			
Regions	Boys	Girls	Difference	>50%	>75%	>90%	>95%
Hong Kong SAR	548 (3.6)	554 (3.4)	-6 (3.6)	1.02	1.08	1.20	1.32
Japan	532 (3.4)	522 (2.9)	10 (3.9)	1.06	1.24	1.55	1.76
South Korea	528 (4.1)	524 (4.0)	4 (5.3)	1.15	1.23	1.43	1.58
Macao SAR	560 (2.2)	556 (2.2)	4 (3.1)	1.06	1.18	1.30	1.35
China (mainland)	597 (2.9)	586 (2.6)	11 (2.4)	1.24	1.40	1.66	1.78
Singapore	571 (1.6)	567 (2.3)	4 (2.3)	1.10	1.21	1.41	1.56
Taiwan	533 (4.3)	529 (4.1)	4 (6.1)	1.07	1.19	1.32	1.36

Notes. Mean performances are from Table II.B1.7.3, Annex B1, PISA 2018 Results (Volume II). Standard errors are in parentheses. Ratio estimates are from the author's own calculation.

acquiring basic capability and a comparatively smaller share of extraordinarily top performers, Macao distinguishes itself from the typical high-performing East Asia systems.

Another characteristic of Macao students' learning performance is its mild level of inequalities. The authors demonstrate this point with two metrics: gender differences and SES-achievement gradients.

Gender Differentials

Gender difference in learning is commonly measured by the mean performance difference between boys and girls. Empirical evidence shows that typically boys perform better than girls in science and math and are more likely to develop a STEM-related career (Sass, 2015; Zhou et al., 2017). As shown in Table 3, data from PISA 2018 indicate that boys have an advantage in math performance over girls among East Asia participating regions, except in Hong Kong where an inverse trend is present. Macao manifests a small gender mean difference. Moreover, unlike Japan and mainland China where boys scored significantly higher than girls, in Macao such gender group difference is not statistically significant, which implies the underlying performance between the two groups is close. In science, which the authors of this chapter did not report due to space limits, the situation is quite similar. Macao reported a small-but-not-significant advantage for girls.

However, gender disparity can manifest in multiple forms. To have a more comprehensive picture, the authors estimated the boy-to-girl ratio at higher tails of performance distribution. The authors use the ratio as a complementary indicator of the gender gap. Again, in the last four columns of Table 3, the authors observe that the gender ratios enlarged gradually when the authors move up the performance ladder. Such a gap consistently exceeds the overall ratios in higher math performance levels in all of these regions, including Hong Kong (from the 75th percentile and above) where girls have a higher mean performance than boys. The increased gender ratio estimates suggest that the mean difference approach misses some important points of the gender disparity story. The gender ratio becomes more prevalent at extremely high-performance levels, indicating that gender gaps are more evident among top performers in math (and science, which the authors did not show here). Macao displays such gaps as well, but its gradients are among the mildest in

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East Asia, not as progressive as those in Japan and mainland China.

Weak link between family background and academic performance

In addition to gender equality, Macao has maintained educational equity with respect to students of different family backgrounds (OECD, 2019b). Results of PISA 2018 suggest that the socioeconomic gradient in Macao was the weakest among all participating countries and regions. Only 1.7% of variance in reading performance was explained by family socioeconomic status, in contrast to the 12% that constituted the OECD average. Similar results were found in mathematics and science (OECD, 2019b, Annex B1.2). Moreover, Macao had the largest proportion of academically resilient students among all participating systems, implying that disadvantaged students in Macao were more likely to overcome social and economic hardship and achieve better learning performance than would be expected.

In a comparative study of trends in socioeconomic achievement gaps, the authors (Lam & Zhou, 2021) investigated the strength of gaps and their development in seven East Asia systems using PISA data from 2003 to 2018. The authors compared the magnitude of gaps between high- and low-, high- and medium-, and medium- and low-SES students in reading, math, and science, indicated by 90/10, 90/50 and 50/10 gaps, respectively. For the "90/10 gap," the authors compared academic performance between those students who stand at the 90th percentile of the socioeconomic distribution with those who stand at the 10th percentile (Reardon, 2011). Their exercise showed that the achievement gaps were statistically smaller in Macao than in the other 6 East Asia systems, regardless of gap types or test domains. For instance, on average, the 90/10 gap estimator in most countries was one standard deviation (SD). Yet the 90/10 gaps in Macao were less than half of one SD, in strong contrast to the particularly large gaps in mainland China.

From a diachronic perspective, Macao has maintained the narrowest gaps (e.g., consistently below 0.5 SD for the 90/10 gaps) since 2003. Recall that the average performance of Macao has improved in the recent decades, while the achievement difference between students from different socioeconomic backgrounds has remained relatively stable and invariably small. This suggests that both affluent and disadvantaged students have acquired progress in learning performance in the past decade.

Further, Lam & Zhou (2021) looked at the performance at the lower tails of socioeconomic distribution and found that students from Macao scored higher than their peers with equivalent backgrounds elsewhere. The edge was most visible at the lowest tiers. When compared with the socioeconomic gradients, the difference between any two adjacent socioeconomic tiers in Macao is the smallest, indicating the least achievement divergence associated with social and economic divisions. In a nutshell, empirical evidence corroborated that Macao has successfully boosted overall educational achievement without leaving the least advantaged behind.

Nurturing STEM talents

As previously mentioned, students in Macao today have the option of participating in a variety of after-school extracurricular activities. Among these, one of the focus areas is STEM (Science, Technology, Engineering, and Mathematics). STEM programs, ranging from enrichment in math and science to robotics competitions, are visible in many local schools. Not only do these programs receive funding, but students who excel at these subjects are also publicly recognized. The Macao Foundation, a government-backed public foundation which aims to promote culture, the arts, science, and education, has set up generous scholarships for STEM subjects. Every primary and secondary student is eligible for the yearly prize. Successful candidates appear at an assembly

Working Paper

held by their school to receive a certificate.

Continued cultivation has sparked students' interests in STEM. The authors compiled Macao's participating record at the International Mathematical Olympiad (IMO) as an illustration. IMO is the world championship mathematics competition for high school students. Figure 3 shows Macao's team results since its first participation in 1990. Macao was not a stellar performer in IMO compared with other East Asia systems such as mainland China, South Korea, or Singapore. But Macao has only missed two years in its thirty-year history and made steady progress regarding global ranking over time. Macao students were also engaged in international competitions such as the International Physics Olympiad (IPhO), the World Robot Olympiad (WRO), and the International Science and Engineering Fair (ISEF) in recent years, drawing growing international attention to this tiny city. Macao's appearance in more and more international competitions reflects its progress in nurturing STEM talents.

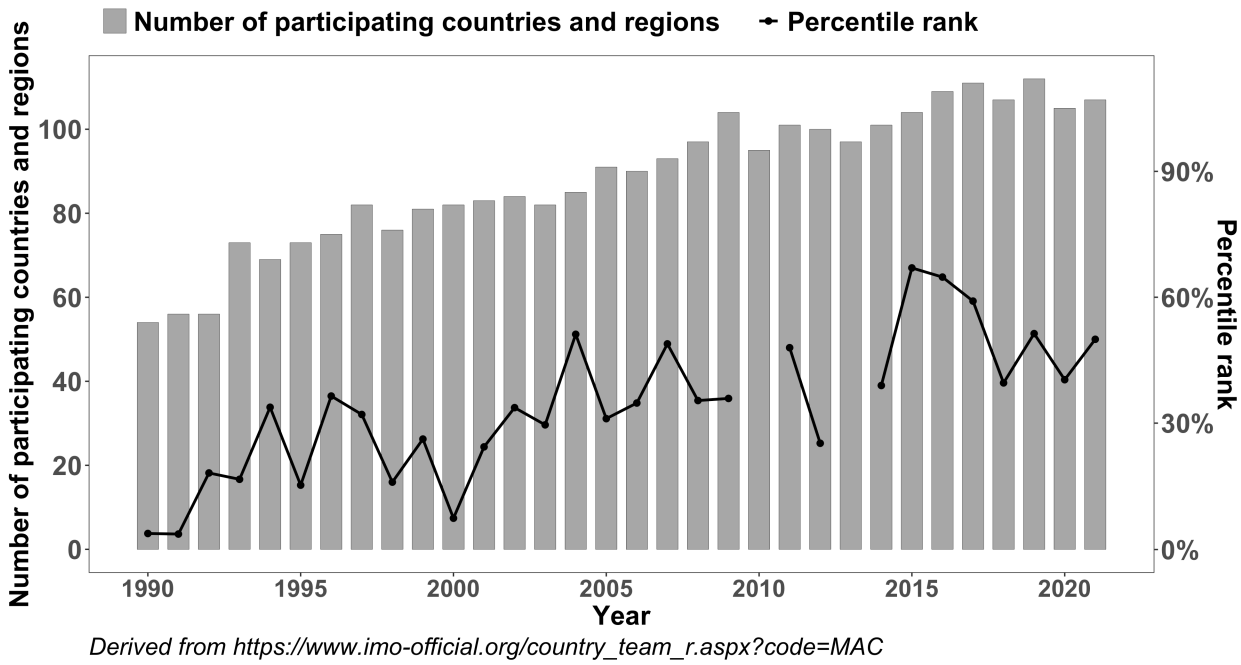

Figure 3: Macao team performance in International Mathematical Olympiad (IMO), 1990-2021

Conclusion

Macao's education system has undergone substantial changes since 1999. As this analysis shows, changes were more prominent in the recent decade. This chapter examined policy efforts in improving the extant education system. It has also outlined some key progress in educational achievement.

Macao's experience was unique in the sense that systemic improvements are anchored by its own political process and a sustained economic boom since the millennium. But the adopted policies by Macao were not unique compared to many other countries engaging in educational reform in the past two decades (Baker, 2014; Ramirez, 2012). The key question is to answer what explains Macao's success story.

The authors argue that Macao's educational success is not simply the result of intensive in-

Work paper

vestment or specific technical treatments. This chapter's explanation has a historical undertone. Macao's story is contingent on several aspects. The first aspect is the fact that Macao's starting point is very low due to its colonial heritage. This historical background – low per-pupil expenditure, disorganized system, and overcrowded facilities – suggests that room for improvement is huge and policy windows are wide. Macao's past left sufficient space for policy experimentation. The second aspect is the timing and the magnitude of the decade-long economic boom. The boom afforded the government with sufficient resources to support wide-spectrum educational projects, as discussed previously. It also means the prevailing human capital view of education, à la crisis rhetoric, did not gain much ground in the local policy discourse.

The third aspect is Macao's peripheral position in the international education community. Macao is a later comer in this regard. It had not attracted any international attention before the PISA exercise. Therefore, combined with the economic success, Macao is free from intensive international or domestic pressure, such as the neoliberal doctrines enacted in many systems, to pursue external agendas. This autonomous policy space allows Macao to draw up its own roadmap. The fourth aspect is social partnership. The current chapter didn't dive deeply into this issue due to space limits, but Macao's success would not bear fruit without the government and schools working closely with one another. The school system is largely made up of mostly private-run institutions. Yet, the relationship between government agencies and schools is more symbiotic than contentious. As a result, observers can often find the government deliberate major reforms initiatives with school leaders. Prioritizing educators' concerns is possible via political arrangements. Macao's political elites, both in Legislative Council and the elite Executive Council, a body of advisors to the Chief Executive of Macao, consist of major representatives from the educational sector. These actors are the hinges to link the education world and the political world (Abbott, 2016, p. 49).

These historical factors coalesced into a re-coupling system focusing on the idea of valuing equity and respecting diversity. Students have universal, rather than restricted, access to quality education, and schools engage in the pursuit of individual potential, rather than standardized ability (Kerckhoff, 2001; Sahlberg, 2007). On one side, expanding public investment in education has greatly expanded access to quality education. Class sizes have become smaller, teacher professionalism has enhanced, and a cohesive curricular framework has been developed. Over 90% of students had access to free or subsidized education, and financial support to school-based development was munificent on top of that. In this way, a network of high-quality schools was established and well accessible to all students. Within the schools, learning assistance was provided for not only students with difficulties but also students with potential in specialized areas. These schools undertook a compensatory role by creating a shared learning experience among the whole student population and targeting resources towards students in need (Dobbie & Fryer, 2009; Downey & Condron, 2016). On the other side, the absence of high-stakes standardized testing protected schools from the market-oriented competition facing many other education systems, and enabled students to fulfill their own potential. This echoes the emphasis on the free individual in the psychological tradition (Miller & Swanson, 1958). Indeed, empirical evidence consistently reveals that high-stakes testing and the accompanying proficiency-based accountability systems jeopardize equality by aggravating the gap between high- and low-performing students (Jennings & Sohn, 2014; Van de Werfhorst & Mijs, 2010). Beyond facilitating academic learning, schools encouraged students to engage in diverse extra-curricular activities. Enrichment activities are well accessible to students even during summer breaks. In the authors' view, all students benefited from the varied learning experience.

These efforts paid off. Macao students have consistently displayed academic excellence in recent international large-scale assessments. Macao has become one of the top performers worldwide. Moreover, the story of Macao is an inclusive one. Even the most disadvantaged students in society

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demonstrated enduring resilience. Disadvantaged students in Macao have better performance than students with equivalent socioeconomic backgrounds elsewhere. Therefore, Macao demonstrated that excellence and equity could and did go hand in hand.

The system has areas for improvement, to be sure. For one thing, the performances of top students were not as good as their counterparts from other East Asia countries and regions. Macao's outstanding average performance was lifted by the high average performance of low-performing students. Whether the leveling up of the bottom was connected to the relatively ordinary performance of high-performing students awaits further investigation.

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